

A Framework for Classifying Kidney Stones Using Explainable Transformer

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Abstract—An AI-driven system for the automatic identification of kidney illnesses has been developed in response to the global nephrologist shortage. Renal failure is a major public health concern. In order to develop an AI-based diagnostic system and support research initiatives in the AI community, such as modeling a digital twin of renal functions, this study focuses on three main categories of renal diseases: kidney stones, cysts, and tumors. It gathers 12,446 annotated CT images of the entire abdomen and urograms.

All classes showed a similar mean color distribution, according to an exploratory data analysis of the collected photos. Furthermore, six different machine learning models were created. Three of these models—EANet [21], DINO, and Swin Transformer [19]—use sophisticated Vision Transformer-based architectures, while the other three are based on well-known deep learning architectures—ResNet50, VGG16, and Vision Transformer (ViT)—that have had their final layers altered.

With an accuracy of 99.37 percent, the Vision Transformer (ViT) outperformed all other models. Transformer-based models perform better than traditional VGG [22], according to comparisons of F1 score, precision, and recall. Additionally, the investigation showed that while ResNet50 performs rather poorly, DINO and VGG16 offer superior localization of anatomical anomalies through Grad-CAM [24] viewing.

We think that kidney tumor, cyst, and stone diagnosis can be greatly aided by the remarkable accuracy of the Vision Transformer (ViT) and the interpretability provided by DINO.

I. INTRODUCTION

Despite current control measures, the prevalence of kidney disease is still rising, making it a serious public health concern. Chronic kidney disease [1], which was the 16th biggest cause of death in 2016 and is expected to increase to the 5th position by 2040, affects more than 10 percent of the world's population. Nephrolithiasis (also called kidney stones) [2], renal cell carcinoma (a type of kidney tumor) [3], and cyst development are the most prevalent kidney diseases that impair kidney function. A kidney cyst is a fluid-filled pocket that develops on the surface of the kidney and is covered by a thin membrane. The kidneys may develop one or more cysts with a water density between 0 and 20 Hounsfield units. About 12 percent of people worldwide suffer from kidney stone disease, which is characterized by the formation of crystalline deposits in the kidneys. Renal cell carcinoma (RCC), sometimes known as kidney cancer, is one of the top 10 most frequent cancers in the world.

In addition to pathology testing, kidney problems are often diagnosed using X-ray, computed tomography (CT), B-ultrasound (US), and magnetic resonance imaging (MRI) machines. Using X-ray beams, the CT scanner analyzes a particular region of the human body to produce a cross-sectional image that provides three-dimensional information on the targeted anatomy [4]. Because CT scans provide high-resolution, slice-by-slice anatomical information, they are very useful for kidney examinations. Renal failure may occur if kidney problems such as cysts, stones, and tumors are not detected and treated very early. Therefore, in order to avoid serious problems, early detection of renal abnormalities is essential.

On the other hand, radiologists and nephrologists are hard to come by. In contrast to Europe, where there are about 25.3 nephrologists per million inhabitants, South Asia has about one nephrologist per million. This discrepancy emphasizes how urgently automated and dependable diagnostic tools are needed.

It is now crucial to create AI-based systems that can correctly detect kidney abnormalities [11] due to the prevalence of kidney-related illnesses, the lack of professionals worldwide, and the quick development of deep learning in medical imaging. Numerous studies have investigated this field [6], [7], but the lack of publicly accessible datasets and the emphasis on binary or limited-class classification tasks continue to be obstacles. Furthermore, a lot of research uses ultrasonic images [5], which offer less structural detail than CT scans.

The "CT KIDNEY DATASET: Normal-Cyst-Tumor and Stone" dataset [13], which comprises 12,446 CT images, was created and annotated for this study. To identify the best architecture for real-time clinical application, we constructed six distinct models and assessed their performance. Additionally, the suggested system helps create a digital twin of kidney function at the pathological level.

To the best of our knowledge, this work is one of the first to thoroughly assess transformer-based models for CT imaging-based multi-class kidney disease classification. The following is a summary of this study's major contributions:

- (i) A comprehensive annotated dataset called "CT KIDNEY DATASET: Normal-Cyst-Tumor and Stone" that includes 12,446 CT scans taken with full abdominal and urogram protocols.
- (ii) Three CNN-based deep learning models (VGG16 and

ResNet50) and a transformer-based Vision Transformer (ViT) model using transfer learning for multi-class renal disease classification are implemented, and Grad-CAM [24] is used for interpretability analysis.

(iii) Three sophisticated transformer-based designs (EAnet [21], Swin Transformer [19], and DINO) are applied for feature representation and classification; DINO is especially used for improved Grad-CAM [24]-based explainability.

(iv) A thorough assessment of every model that shows the superiority of transformer-based methods utilizing performance metrics like accuracy, precision, recall, F1 score, and confusion matrix.

This is how the rest of the paper is organized. Background research on deep learning for renal disease diagnosis is presented in Section II. The approach, including data pretreatment, model designs, and evaluation techniques, is covered in Section III. The experimental results and analysis are covered in Section IV, and the study is concluded in Section V.

II. BACKGROUND STUDY

Research on automated medical diagnosis has significantly increased due to the quick development of deep learning and its extensive applications in image analysis and classification [10], [11], especially in radiological image classification and segmentation tasks. In order to accurately diagnose and detect a wide range of disorders, deep learning approaches have shown amazing effectiveness in extracting useful features from complex medical imaging data. Because they can learn hierarchical feature representations, convolutional neural networks (CNNs) like ResNet, EfficientNet [17], and Xception [16] have become quite popular in classification problems that use transfer learning. In order to detect anomalies in medical images, these architectures successfully capture local spatial patterns including edges, textures, and forms.

In medical imaging applications, where large annotated datasets are frequently hard to come by, transfer learning is essential. Transfer learning makes it possible to reuse learnt feature representations and modify them for domain-specific tasks by utilizing models that have already been trained on extensive datasets like ImageNet. This method enhances performance and drastically cuts down on training time, particularly when working with small medical datasets. However, because of their confined receptive fields, CNN-based models are intrinsically unable to capture long-range dependencies, which may limit their capacity to represent global contextual interactions inside intricate anatomical structures.

Transformer-based architectures, which were first created for natural language processing, have been effectively modified for computer vision tasks in order to get over these restrictions. These models are especially well-suited for medical imaging, where contextual information is crucial, because they rely on self-attention mechanisms that allow them to collect global dependencies throughout the entire image. By interpreting images as patch sequences and using transformer encoders to learn global representations, the Vision Transformer (ViT) [18] brought about a paradigm shift. By adding effective

attention techniques that cut down on redundancy and boost computational performance, variations like External Attention Transformer (EAnet [21]) further increase this capability. To reconcile local and global feature extraction, hierarchical transformer models like Swin Transformer [19] have also been developed.

Self-supervised learning techniques like DINO have become effective ways for representation learning, in addition to supervised learning approaches. DINO is especially useful in medical areas since it allows models to learn significant visual features without requiring large amounts of labeled data. These acquired representations enhance interpretability and classification performance and are very transportable. By emphasizing clinically significant areas in medical images, DINO-based models offer improved insight into the decision-making process when paired with visualization techniques like Gradient-weighted Class Activation Mapping (Grad-CAM [24]). This increases confidence and dependability in AI-assisted diagnosis.

Several deep learning methods have been investigated for the classification of renal diseases. In one study, morphological operations, median filtering, and Gaussian filtering were used to improve renal ultrasound pictures. Principal Component Analysis (PCA) was then used to extract features, and K-Nearest Neighbors (KNN) was used for classification [12]. With a maximum F1 score of 0.853, another study compared CNN-based methods with conventional machine learning algorithms like Decision Trees (DT), Random Forest (RF), Support Vector Machines (SVM), Multilayer Perceptron (MLP), and Naive Bayes.

In a different method, features were extracted from ultrasound pictures using pre-trained deep neural networks including ResNet-101, ShuffleNet, and MobileNet-v2. SVM with a majority voting mechanism was then used for classification, yielding an accuracy of 95.58 percent. deep networks with regularization [23] with residual dual-attention modules (RDA) have been developed for CT-based renal cyst segmentation [8], [9]. While multi-stage CNN pipelines have been designed to detect kidney stones and urinary tract problems with almost 95 percent accuracy, other studies have investigated hybrid techniques that mix handmade characteristics with deep learning representations.

Despite these developments, the majority of current research focuses mostly on X-ray or ultrasound imaging modalities, which offer less anatomical information than CT imaging. Furthermore, rather than tackling multi-class classification problems including multiple renal diseases, many efforts are limited to binary classification tasks, such as identifying the presence or absence of a disease. CT imaging provides a more thorough image of kidney structures and is therefore more appropriate for differentiating between complicated disorders including cysts, tumors, and stones due to its high spatial resolution and extensive cross-sectional information.

We created an extensive CT-based dataset with kidney cysts, tumors, stones, and normal patients in order to overcome these constraints. For categorization in this study, we used deep

learning architectures such as VGG16, ResNet50, and Vision Transformer (ViT). Moreover, sophisticated transformer-based models like self-supervised DINO and EANet [21] were used to improve interpretability and feature learning. Better model decision region visualization is made possible by the integration of DINO with Grad-CAM [24], which also increases the system's transparency and offers clinically significant explanations.

Overall, our method uses both transformer-based and convolutional architectures to accomplish reliable and precise multi-class kidney disease classification using CT imaging. This work fills important research gaps and advances the creation of trustworthy AI-assisted diagnostic systems for practical clinical applications by fusing explainability methodologies

with high-performance models.

III. METHODOLOGY

In order to create a database for results pertaining to kidney stones, tumors, normals, and cysts, we first collected and annotated datasets. Image scaling, normalization [25], and stratified data separation are among the preprocessing methods used. We then employed six models to assess our dataset, including DINO, VGG16, ResNet50, and three Vision Transformer variations (EANet [21], Vision Transformer (ViT) [18], and Swin Transformer [19]). Test data that had never been seen before was used to assess the models' performance. Fig. 1 shows specifics about the entire experimental pipeline.

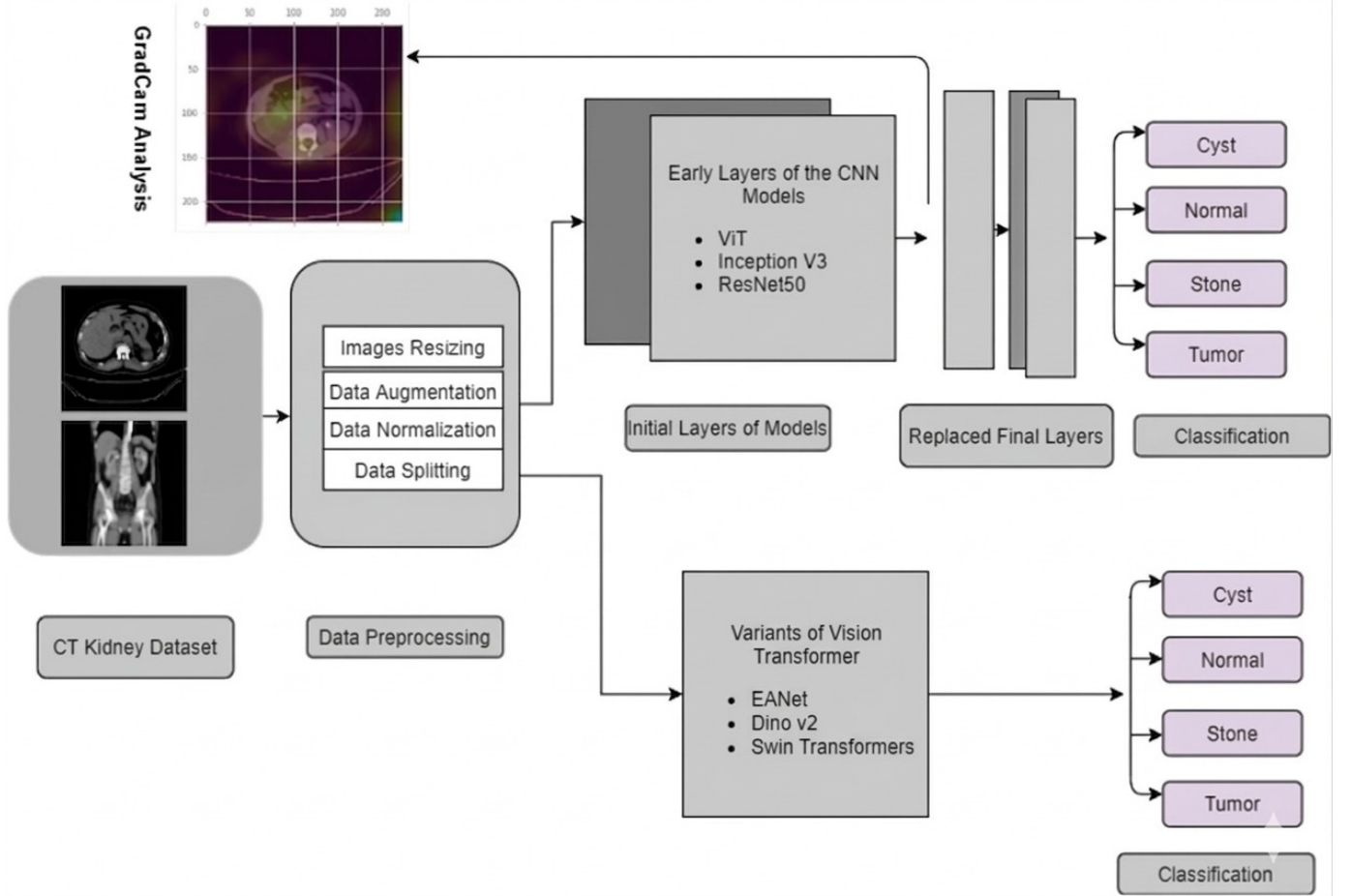


Fig. 1: Complete pipeline of the proposed framework including preprocessing, model training, and Grad-CAM [24] based explainability using DINO and transformer models

The following subsections comprise the methodology: model designs, evaluation methodologies, preprocessing methods, and dataset description.

A. Dataset Description

The Picture Archiving and Communication System (PACS) and workstations at a hospital in Dhaka, Bangladesh, where patients had previously been diagnosed with kidney cancers,

cysts, normal conditions, or stones, provided the dataset. Before being included in the study, each subject gave their informed consent. The regulatory authorities at Dhaka Central International Medical College and Hospital (DCIMCH) approved the data gathering process, and all procedures adhered to acceptable ethical standards.

The entire abdominal and urogram process was followed for choosing coronal and axial slices from contrast-enhanced and

non-contrast CT examinations. Regions of interest were extracted for each class after the DICOM data was meticulously vetted to separate specific diagnoses. The photos were transformed into a lossless JPEG format and all patient-identifiable information and metadata were eliminated to protect privacy.

The Philips IntelliSpace Portal 9.034 application, which offers sophisticated visualization features for radiological investigation, was used to annotate the data. Additionally, picture

conversion and inspection were done using the Sante DICOM Editor. To guarantee labeling accuracy, a doctor and a medical technologist examined each annotated image in more detail.

12,446 photos make up the final collection, which includes 3,709 cyst cases, 5,077 normal cases, 1,377 stone cases, and 2,283 tumor cases. To enable repeatability and additional research, the dataset has been made publically available on Kaggle [13].

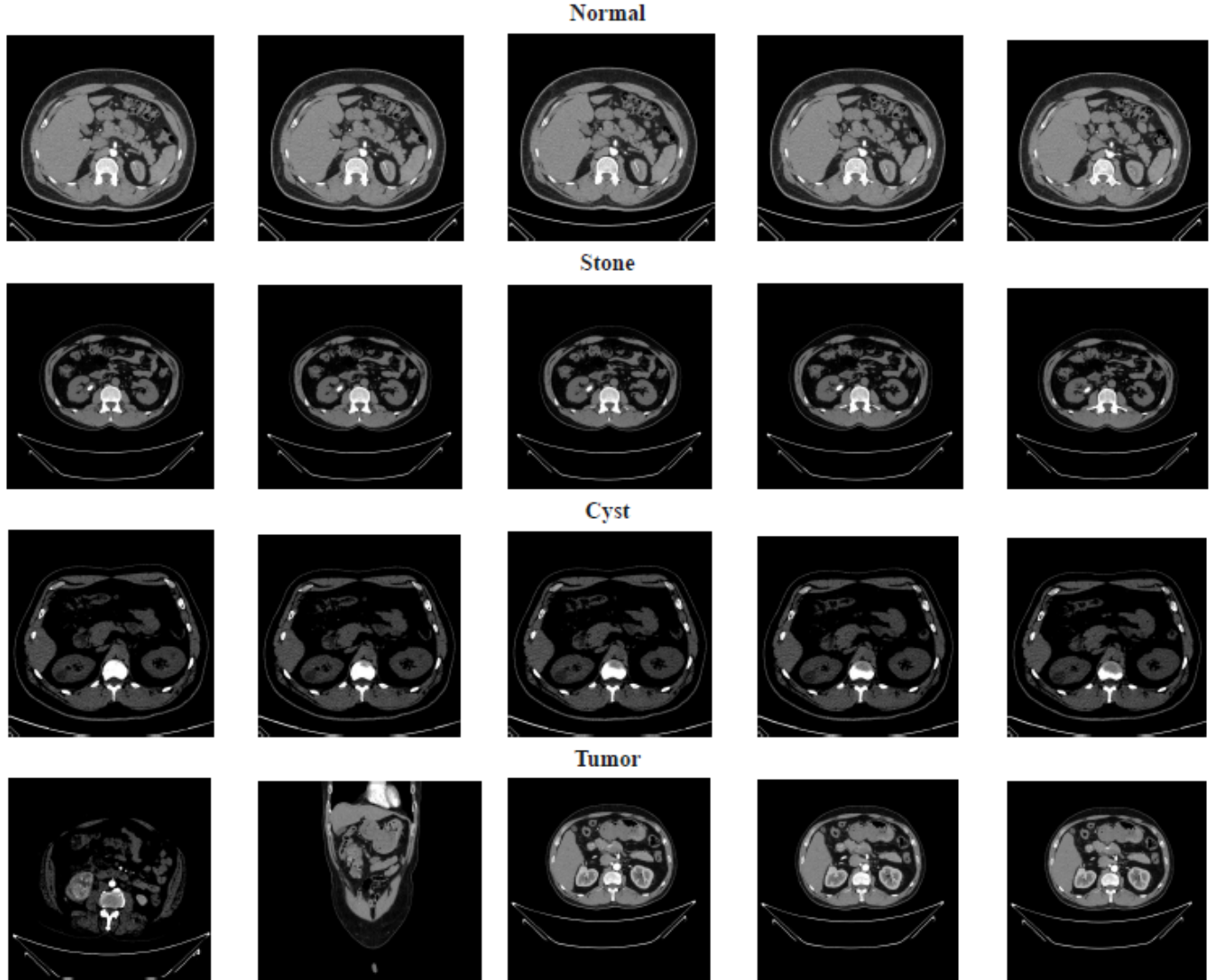


Fig. 2: Representative CT images of kidney cyst, normal, stone, and tumor classes with highlighted regions of interest

The distribution of mean pixel intensity levels within each class and throughout the dataset is shown in Figures 3 and 4. The distributions show that all classes have comparable statistical traits. Furthermore, the majority of samples are clustered around central values, with minor changes shown in the stone and cyst classes, according to the plot of mean against standard deviation (Fig. 5).

It is not possible to differentiate between cysts, tumors, and stones using only basic statistical criteria due to the

substantial overlap in statistical distributions between classes. This emphasizes how important it is to use transformer-based and deep learning models that can extract intricate spatial and contextual information from CT scans.

B. Dataset Acquisition and Initial Structural Organization

1) *Directory-Based Data Collection:* A hierarchical directory structure was used to store the raw dataset on Google

Drive, with each top-level folder representing a distinct diagnostic class: *Cyst*, *Normal*, *Stone*, and *Tumor*.

This file system structure has to be traversed in order to create a programmatically available dataset in the first preprocessing stage. This was accomplished by iterating through each class directory using Python's `os.listdir()` function. Two important characteristics were noted for each image file: the class label (*label*) and the absolute file path (*image_path*). After being kept in parallel lists, these characteristics were merged into a structured dataset.

2) *Construction of the Master DataFrame*: A Pandas DataFrame, which acts as the main data structure for the entire pipeline, was created by combining the gathered file paths and labels. Effective data processing, filtering, and transformation are made possible by the DataFrame's organized and tabular representation of the dataset.

A DataFrame has a number of benefits over raw lists or NumPy arrays, such as labeled columns, boolean indexing support, and easy integration with machine learning libraries like *scikit-learn* and *imbalanced-learn*. The final DataFrame, which has the columns *image_path* and *label*, serves as the single source of truth for all subsequent operations, such as batch generation, stratified splitting, label encoding, and oversampling.

C. Data Quality Verification and Integrity Assurance

1) *Duplicate Detection*: Finding duplicate entries with `df.duplicated().sum()` was a crucial data quality validation step. Duplicate samples, which result from repetitive file inclusion or inconsistent file systems, can cause serious evaluation bias in medical imaging workflows.

The model may memorize identical images from both training and testing sets, leading to performance measures that are artificially inflated. The reliability of model evaluation is compromised by this problem, which is referred to as *data leakage* [14].

Duplicate detection was carried out before dataset splitting in order to reduce this danger. A fair evaluation of model generalization is ensured and statistical independence is maintained by making sure the training, validation, and test sets are mutually exclusive.

2) *Missing Value Inspection*: Using `df.isnull().sum()`, the dataset was further verified for missing or null items. In a path-label DataFrame, unlabeled samples are represented by null values in *label*, whereas missing or unavailable image files are indicated by null values in *image_path*. Runtime errors during data loading and training may result from such discrepancies.

Early detection of missing values guarantees robustness and stability in the training pipeline by enabling suitable corrective actions, such as eliminating corrupted entries or carrying out manual verification.

3) *Distributional Visualization and Class Imbalance Diagnosis*: A custom visualization function `visualize_label_distribution()` was used to create a pie chart and a count plot (bar chart) in order to

assess class distribution. Both absolute and relative viewpoints on class frequencies are offered by these complementary graphics.

The count plot makes it easier to identify majority and minority classes by providing an accurate numerical representation of each class. By showing the percentage contribution of each class to the total dataset, the pie chart, on the other hand, illustrates proportional distribution.

Due to the disproportionate incidence of clinical illnesses, class imbalance is a common problem in medical imaging datasets [11]. Standard loss functions like cross-entropy have a tendency to skew the model in favor of majority classes when training on unbalanced data. Because of this, the model can have a high overall accuracy yet be unable to accurately identify minority classes.

Such behavior is prohibited in medical diagnosis because it can have serious clinical repercussions to miss important disorders (like cancers). As a result, a critical stage in the model creation process is recognizing and resolving class imbalance.

D. Label Encoding: From Categorical Strings to Ordinal Integers

1) *Theoretical Basis*: Because deep learning models work with numerical tensors [25], categorical labels like *Cyst*, *Normal*, *Stone*, and *Tumor* need to be transformed into numerical form. Using `LabelEncoder` from the *scikit-learn* library, which maps each class to a distinct integer in the range $[0, N - 1]$, where N is the number of classes, label encoding was carried out.

The resulting mapping is shown in Table I.

TABLE I: Label Encoding Scheme

Original Label	Encoded Integer	Assignment Rule
Cyst	0	Alphabetical order
Normal	1	Alphabetical order
Stone	2	Alphabetical order
Tumor	3	Alphabetical order

A new column called *category_encoded* was used to store the encoded labels. To guarantee label mappings were consistent throughout training, validation, and test sets, the `fit_transform()` operation was used before dataset splitting. Only numerical representations were kept for model training once the initial categorical label column was eliminated.

2) *Why Ordinal Encoding Instead of One-Hot Encoding?*: Because ordinal integer encoding is compatible with the loss function employed in this investigation, it was selected in place of one-hot encoding. There is no need to translate labels into high-dimensional one-hot vectors because sparse categorical cross-entropy accepts integer class labels directly.

This method maintains mathematical equivalency to categorical cross-entropy while drastically reducing computing cost and memory overhead, particularly for large datasets. For multi-class classification applications, ordinal encoding thus offers a more effective and scalable method.

subsubsectionString Casting for Keras Flow-from-DataFrame Compatibility

Casting integer-encoded labels back to string format before sending them to Keras’s `flow_from_dataframe()` generator was a small but crucial technical detail. This may seem unnecessary, but it guarantees compatibility with the generator’s internal operation.

The generator expects labels to be either strings or string representations of integers when `class_mode='sparse'` is used. To translate these labels into numerical form, it internally creates a `class_indices` mapping. We guarantee that this mapping is deterministic and constant throughout training, validation, and test generations by transforming encoded labels into string format (e.g., ‘0’, ‘1’, ‘2’, ‘3’).

By preventing inadvertent label reordering or dataset mismatch, this stage ensures accurate classification metric computation.

E. Stratified Dataset Partitioning

1) *The Train-Validation-Test Paradigm:* Three mutually exclusive subsets of the dataset were created: training (80%), validation (10%), and test (10%). The conventional supervised learning methodology is followed in this partitioning.

Backpropagation is performed to optimize model parameters using the training set. Hyperparameter adjustment and early halting are guided by the validation set, which is also used to track training results. In order to provide an objective assessment of model generalization, the test set is used only for the final performance evaluation and is not visible at all during training.

2) *Stratified Splitting and Its Statistical Necessity:* `train_test_split` from *scikit-learn* was used to split the dataset, and the `stratify` parameter was applied to the encoded labels. Each subset is guaranteed to retain about the same class distribution as the original dataset thanks to stratification.

Random sampling may lead to disproportionate class representation in the absence of stratification, especially for minority classes. This could result in underrepresentation or absence of important classes (such as tumors) in validation or test sets in medical imaging, leading to inaccurate performance estimations.

The training set and a temporary holdout set (20%) were produced by the first split (`train_size=0.8`). Using `test_size=0.5`, the holdout set was then split equally into validation and test sets, yielding 10% for each. To guarantee reproducibility, a fixed `random_state=42` was employed.

TABLE II: Dataset Partitioning Strategy

Subset	Proportion	Purpose	Shuffle
Training	80%	Model training	Yes
Validation	10%	Hyperparameter tuning	Yes
Test	10%	Final evaluation	No

`shuffle=False` was purposefully utilized for the test generator in order to maintain sample order. Confusion matrices and classification metrics can be computed accurately

since this guarantees alignment between anticipated outputs and ground truth labels.

F. Class Imbalance Remediation via Random Oversampling

1) *The Problem of Class Imbalance in Deep Learning:* A major problem in medical imaging datasets is class imbalance, when some illnesses are less common than others. Models are biased toward majority classes because standard loss functions, like cross-entropy, treat all samples equally.

Consequently, models may have low recall for minority groups but excellent overall accuracy. This is especially problematic in medical diagnosis since it can have serious repercussions if uncommon but important illnesses (like cancers) are missed.

Therefore, in order to guarantee balanced learning and trustworthy clinical performance, class imbalance must be addressed.

2) *Random Oversampling: Mechanism and Justification:* Using the `RandomOverSampler` from the *imbalanced-learn* package, Random Oversampling was used to correct class imbalance. By randomly replicating preexisting samples until all classes have an equal distribution, this method improves the representation of minority classes.

In order to guarantee that every class contributes evenly to the loss function during optimization, the training dataset must be balanced. This encourages more stable and balanced learning and keeps the model from favoring majority courses.

Crucially, oversampling was used **only to the training set**. To maintain their initial class distributions, the validation and test sets were left unaltered. This design decision avoids artificially inflated evaluation findings and guarantees that performance measurements match real-world data properties.

3) *Why Not Undersampling or SMOTE?:* Alternative techniques including the Synthetic Minority Oversampling Technique (SMOTE) and random undersampling were taken into consideration. Because random undersampling eliminates samples from majority classes and may result in the loss of important data, it was not used.

SMOTE is less appropriate for image datasets saved as file paths, even though it works well for structured numerical data. It necessitates feature-space interpolation, which discards spatial information and increases computing complexity by flattening images into vectors.

Random Oversampling, on the other hand, effectively integrates with the image data generator pipeline while maintaining image integrity by working directly at the DataFrame level.

4) *Implications for Training Dynamics:* Gradient updates are prevented from being dominated by any one class by using a balanced training dataset. This increases the model’s capacity to learn representative features for every class and improves convergence stability.

The resulting balanced dataset, called `train_df_balanced`, is utilized only for training; test and validation datasets are kept in their original distributions for objective performance assessment.

G. Image Loading, Decoding, and Preprocessing via Keras ImageDataGenerator

1) *Architecture of the Image Loading Pipeline:* TensorFlow/Keras's `ImageDataGenerator` and the `flow_from_dataframe()` technique were used to implement image loading. Instead of fully loading photos into memory during training, this pipeline uses a lazy loading method, which reads, decodes, resizes, and processes images on the fly.

Medical imaging files, which are usually huge and high-resolution and may surpass available memory resources if preloaded, are especially well suited for this method.

2) *Pixel Value Rescaling:* `rescale=1./255` was used to normalize all photos, changing pixel intensity values from the range $[0, 255]$ to $[0, 1]$. This normalization [25] guarantees that the model functions within a constant input range and enhances numerical stability throughout training.

Large pixel values may cause unstable gradients, delayed convergence, and possibly activation function saturation if they are not properly normalized.

3) *Image Resizing:* To guarantee uniform input dimensions throughout the dataset, every image was shrunk to a fixed resolution of 224×224 pixels using `target_size=(224, 224)`. Compatibility with deep learning models and effective batching depend on this standardization.

The selected resolution is consistent with conventional Vision Transformer (ViT) [18] settings, which divide images into processing-ready fixed-size patches.

4) *Color Mode and Channel Handling:* `color_mode='rgb'` was used to import images, producing three-channel tensors of shape $(224, 224, 3)$. In order to preserve compatibility with pre-trained deep learning architectures, which usually anticipate multi-channel inputs, CT images—which are by nature grayscale—were recorded in RGB format.

5) *Batch Size Selection:* In order to balance training stability and computational efficiency, a batch size of 16 was used. While bigger batch sizes necessitate more memory and may worsen generalization performance, smaller batch sizes add stochasticity to gradient updates, which can enhance generalization.

H. Deliberate Absence of Online Image Augmentation

No stochastic data augmentation methods (such as rotation, flipping, or brightness modulation) were used, in contrast to traditional computer vision pipelines.

Medical imaging domain-specific limitations are the driving force behind this choice. In CT images, anatomical features exhibit consistent orientations and intensity distributions that are clinically significant. These features could be distorted and unnatural patterns introduced by arbitrary changes, which could impair model performance.

Instead, oversampling techniques were used to solve class imbalance, guaranteeing balanced learning without changing the inherent characteristics of medical images.

I. Model Architecture: DINOv2-Inspired Vision Transformer

1) *Overview:* The Vision Transformer (ViT) [18] architecture, which was influenced by DINOv2, serves as the foundation for the suggested classification model. The model uses self-attention mechanisms to extract global contextual information and long-range dependencies from images that are processed as patch sequences.

Improved interpretability and robustness are made possible by DINO-based feature learning, especially when combined with Grad-CAM [24] for visual explanation.

2) *Patch Embedding:* Each 224×224 input picture is split into 16×16 non-overlapping patches, totaling 196 patches. A convolutional projection layer is used to linearly project each patch into a fixed-dimensional embedding space, creating a series of patch embeddings.

3) *Class Token and Positional Encoding:* The series of patch embeddings is followed by a learnable classification (CLS) token. This token acts as the representative feature for categorization by combining global data from all patches.

The model retains information about patch placements by adding positional embeddings to the patch embeddings in order to maintain spatial linkages.

4) *Transformer Encoder Blocks:* The model consists of multiple Transformer encoder layers, each comprising:

- Multi-head self-attention for modeling global dependencies
- Layer normalization [25] for training stability
- Feed-forward neural networks (MLP) with GELU activation
- Residual connections to facilitate gradient propagation

These components enable the model to effectively capture both local and global patterns within CT images.

5) *Classification Head:* To generate probability distributions across the four classes—Cyst, Normal, Stone, and Tumor—the output corresponding to the CLS token is retrieved and fed through a fully connected layer with softmax activation.

Sparse categorical cross-entropy, which works with integer-encoded labels, is used to train the model.

J. Optimization, Regularization, and Training Strategy

1) *Adam Optimizer with Low Learning Rate:* The Adam optimizer was used to optimize the model at a learning rate of 1×10^{-5} . Adam is a good fit for training deep transformer-based architectures since it calculates adaptive learning rates using first- and second-order gradient moments.

Because transformer models are sensitive to significant changes in attention weights, a low learning rate was used to guarantee steady convergence. For medical imaging tasks in particular, this option strikes a balance between training stability and convergence speed.

2) *Early Stopping:* Validation loss (`val_loss`) was tracked using an `EarlyStopping` callback with a patience of five epochs. In order to avoid overfitting, training was stopped when the validation loss did not improve over successive epochs.

In order to increase generalization performance, the parameter `restore_best_weights=True` makes sure that the model keeps the weights that correspond to the lowest validation loss instead of those from the last training epoch.

3) *Model Checkpointing*: Every time validation performance increased, model weights were saved via a `ModelCheckpoint` callback. This ensures that the top-performing model is maintained throughout training and can be recovered without retraining.

In cloud-based or resource-constrained contexts, where training processes could be stopped, checkpointing is especially helpful.

4) *Two-Phase Training Strategy*: There were two consecutive parts to the training. In the first stage, an initial optimal set of weights was obtained by training the model with early halting. Training was restarted in the second phase once these optimal weights were loaded into a fresh model instance.

By using this technique, the optimizer can reset its internal states, which helps the model avoid possible local minima and enhance convergence.

K. Explainability via Grad-CAM [24] on Transformer Features

1) *Motivation for Explainability*: Interpretability is essential for clinical uptake and trust in medical AI systems. Without supporting data, black-box projections are insufficient for practical implementation. Clinicians can confirm if predictions are based on pertinent anatomical structures by using explainability techniques, which offer visual explanation for model conclusions.

2) *Grad-CAM [24] Adaptation for Vision Transformer and DINO*: Gradient-weighted Class Activation Mapping (Grad-CAM [24]) was modified for transformer-based architectures, such as Vision Transformer (ViT) [18] and DINO-based models, to improve interpretability.

Transformer models interpret images as sequences of patch embeddings, in contrast to convolutional neural networks, which rely on spatial feature maps. Gradients are therefore calculated in relation to the final transformer encoder layer’s output tokens.

Importance weights are derived by calculating the gradient of the predicted class score with respect to each token embedding. A relevance map is then created by combining these weights with the matching token representations.

The resulting attention map was upsampled to match the original image resolution of 224×224 after being molded into a 14×14 grid, which corresponded to the 196 image patches.

Self-supervised attention maps offer extra semantic localization for DINO-based representations, improving the quality of Grad-CAM [24] displays. This combination makes it possible to identify clinically significant areas—like kidney stones, cysts, and tumors—more accurately.

3) *Visualization and Clinical Interpretation*: To show the areas that most strongly influence the model’s predictions, the Grad-CAM [24] heatmaps are superimposed over the original

CT scans. To gain a better understanding of model behavior and decision-making tendencies, both properly and wrongly classified samples are examined.

Verification of the model’s concentration on therapeutically relevant anatomical regions—such as kidney structures linked to cysts, stones, and tumors—instead of unimportant background features is made possible by this visualization approach. The suggested system’s interpretability and clinical reliability are improved by the incorporation of transformer-based attention and DINO-driven representations, which further improve localization quality.

L. End-to-End Data Pipeline Summary

Table III summarizes the whole preprocessing and training pipeline, describing each step and its associated goal.

TABLE III: End-to-End Data Pipeline

Step	Operation	Purpose
1	File traversal	Structured dataset creation
2	Duplicate removal	Prevent data leakage
3	Null value check	Ensure data integrity
4	Distribution analysis	Detect class imbalance
5	Label encoding	Convert labels to numeric form
6	Stratified splitting	Maintain class proportions
7	Oversampling	Balance training dataset
8	Type casting	Ensure generator compatibility
9	Rescaling	Normalize pixel values
10	Image resizing	Standardize input dimensions
11	RGB conversion	Match model input requirements
12	Batch generation	Efficient training pipeline

M. Image Processing

All DICOM images were standardized to satisfy the deep learning models’ input criteria after being converted into JPG format. A uniform resolution of 224×224 pixels was used for all models, in contrast to previous heterogeneous resizing techniques, to provide uniformity and compatibility with Vision Transformer (ViT) architectures.

In order to guarantee equal class representation during training, the dataset was shuffled and balanced by choosing 1,300 samples from each class (Cyst, Normal, Stone, and Tumor). Pixel value normalization [25] was a basic preprocessing step. Z-normalization [25] [25] was also used, as demonstrated in (1):

$$\hat{X} = \frac{X[:, i] - \mu_i}{\sigma_i} \quad (1)$$

where μ_i represents the mean and σ_i represents the standard deviation of the feature distribution.

Eighty percent of the dataset was used to train each model; the remaining portion was set aside for testing and validation. Google Colab Pro with 26.3 GB RAM and 16 GB GPU memory (CUDA 11.2) was used for the training. Training was carried out for up to 100 epochs with early halting, using a batch size of 16.

N. Deep Learning Models

Six models, including both convolutional and transformer-based architectures, were used in this investigation. VGG16, ResNet50, EANet [21], Swin Transformer [19], Vision Transformer (ViT) [18], and a DINO-based self-supervised model for improved feature representation and Grad-CAM [24] visualization are among the models.

1) *VGG16*: A transfer learning strategy was utilized to modify the VGG16 model [22], a popular convolutional neural network. The initial fully connected layers were swapped out with a specially designed classification head, but the convolutional basis of the pre-trained model was kept to take advantage of its capacity to extract hierarchical spatial data.

The altered architecture consists of a flatten layer, a dense layer with ReLU activation, a dropout layer for regularization, and global average pooling to minimize spatial dimensions. The last layer performs four-class classification (Cyst, Normal, Stone, Tumor) using softmax activation.

Of the 14,747,780 parameters in the improved VGG16 model, 4,752,708 are trainable and 9,995,072 are not. This arrangement lowers the possibility of overfitting while facilitating effective feature extraction.

2) *ResNet50*: ResNet50 [14] uses residual (skip) connections, which enable gradient flow during backpropagation, to overcome the degradation issue in deep neural networks. The model may learn identity mappings thanks to these connections, which enhances convergence and deep architecture performance.

In this study, a task-specific dense network appropriate for four-class kidney disease classification was used to replace the final classification layers of a pre-trained ResNet50 model. The top layers were trained on the target dataset, but the feature extraction layers were mostly frozen to maintain learnt representations.

With 23,719,108 parameters—135,492 trainable and 23,583,616 non-trainable—the updated ResNet50 model strikes a balance between computational efficiency and performance.

3) *Vision Transformer (ViT)*: By representing images as patch sequences, the Vision Transformer (ViT) [18] marks a paradigm change from convolution-based systems. Fixed-size patches are created from each input image and linearly embedded into a high-dimensional feature space.

The resulting sequence is processed via several transformer encoder layers after positional encodings are added to preserve spatial information. Multi-head self-attention and feed-forward neural networks make up each encoder layer, allowing the model to capture global contextual linkages and long-range dependencies.

A learnable classification (CLS) token is appended to the sequence and serves as the aggregated representation for final prediction. Class probabilities are produced by passing the CLS token's output through a fully connected layer with softmax activation.

In medical imaging activities, where global contextual awareness is essential for identifying minute anatomical differences, ViT is very helpful.

4) *DINO-Based Model*: A self-supervised learning system based on DINO (Distillation with No Labels) was added to further improve feature representation. In contrast to conventional supervised methods, DINO learns meaningful visual representations without the need for labeled data, which makes it ideal for medical imaging applications where annotation is expensive and scarce.

DINO employs a teacher-student architecture in which the student network learns to match the output distributions of a momentum-updated teacher network. This procedure makes it possible to extract semantically rich and invariant information from CT scans.

In this work, DINO is coupled with Grad-CAM [24] for explainability in addition to being used for feature learning. The attention maps constructed using DINO-based features enable more precise and semantically meaningful localization of disease-relevant regions compared to standard CNN-based techniques. This greatly increases the model's interpretability and boosts clinical confidence in it.

5) *Transformer-Based Models: External Attention Transformer (EANet [21])*:

In order to increase efficiency and decrease redundancy in feature representations, EANet [21] adds an external attention mechanism to the standard Vision Transformer [18]. EANet uses two shared learnable memory units, M_k and M_v , to compute attention rather than only within the input sequence.

The external attention mechanism is defined as:

$$\Lambda = \text{Norm}(FM_k^T) \quad (2)$$

$$F_{\text{out}} = \Lambda M_v \quad (3)$$

This approach permits efficient calculation of attention while preserving global contextual information. The revised feature representations are subsequently put through a classification head to provide probability scores for renal disease categories.

6) *Swin Transformer [19]*: A shifting window-based self-attention mechanism is introduced by the Swin Transformer [19], a hierarchical Vision transformer that strikes a balance between representational power and computing efficiency.

The input image is first divided into non-overlapping patches by the model, and then linear embedding is applied. Each of the architecture's numerous stages is made out of a number of Swin Transformer blocks. These blocks include multi-layer perceptrons (MLPs), layer normalization [25] (LN), and multi-head self-attention (MSA).

The Swin Transformer's shifting window system, which permits cross-window connections and supports both local and global feature learning, is its primary innovation. The model successfully captures spatial interactions at various scales because to its hierarchical nature.

The Swin Transformer [19] is used in this work to classify kidney CT pictures into multiple classes, and it performs well in recognizing cysts, tumors, stones, and normal instances.

TABLE IV: Number of Parameters of Different Models

Model	Total Parameters	Trainable Parameters
VGG16	14,747,780	4,752,708
ResNet50	23,719,108	135,492
EANet [21]	600,907	600,900
Swin Transformer [19]	412,788	396,372
ViT	50,688	50,688
DINO	76,032	76,032

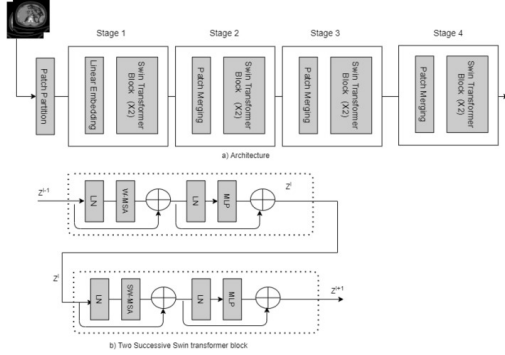


Fig. 3: Architecture of Swin Transformer [19] used in the study

O. Performance Evaluation Methods

Standard classification criteria, such as accuracy, precision, sensitivity (recall), and F1-score, are used to assess the performance of the suggested models. True positives (TP), false

positives (FP), true negatives (TN), and false negatives (FN) are the components of the confusion matrix from which these metrics are formed.

While sensitivity (recall) indicates the model's capacity to accurately detect positive cases, which is especially crucial in medical diagnosis, accuracy quantifies the total correctness of predictions. The accuracy of positive predictions is indicated by precision (positive predictive value, or PPV). The F1-score offers a fair assessment of the model's performance by calculating the harmonic mean of precision and recall.

The following is a definition of the evaluation metrics:

$$\text{Accuracy}_i = \frac{\text{TP}_i + \text{TN}_i}{\text{TP}_i + \text{TN}_i + \text{FP}_i + \text{FN}_i} \times 100\% \quad (4)$$

$$\text{Precision}_i = \frac{\text{TP}_i}{\text{TP}_i + \text{FP}_i} \quad (5)$$

$$\text{Sensitivity}_i = \frac{\text{TP}_i}{\text{TP}_i + \text{FN}_i} \quad (6)$$

$$\text{F1_score}_i = 2 \times \frac{\text{Precision}_i \times \text{Sensitivity}_i}{\text{Precision}_i + \text{Sensitivity}_i} \quad (7)$$

where $i \in \{\text{Cyst, Normal, Stone, Tumor}\}$ denotes the class label.

In order to assess classification performance across various thresholds, Receiver Operating Characteristic (ROC) curves are also created. The Area Under the Curve (AUC) is utilized as a summary indicator, where higher AUC values imply greater discriminative power of the model.

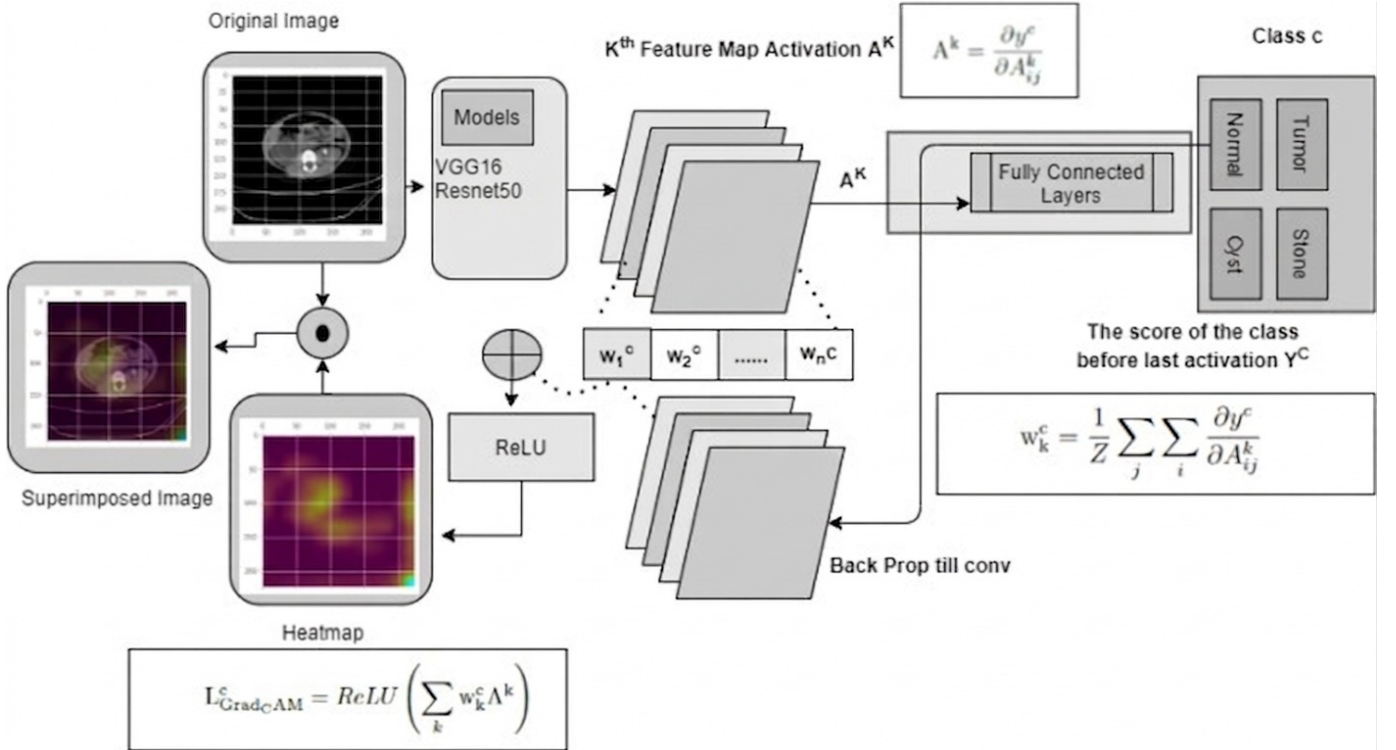


Fig. 4: Sample CT images of kidney classes: cyst, normal, stone, and tumor

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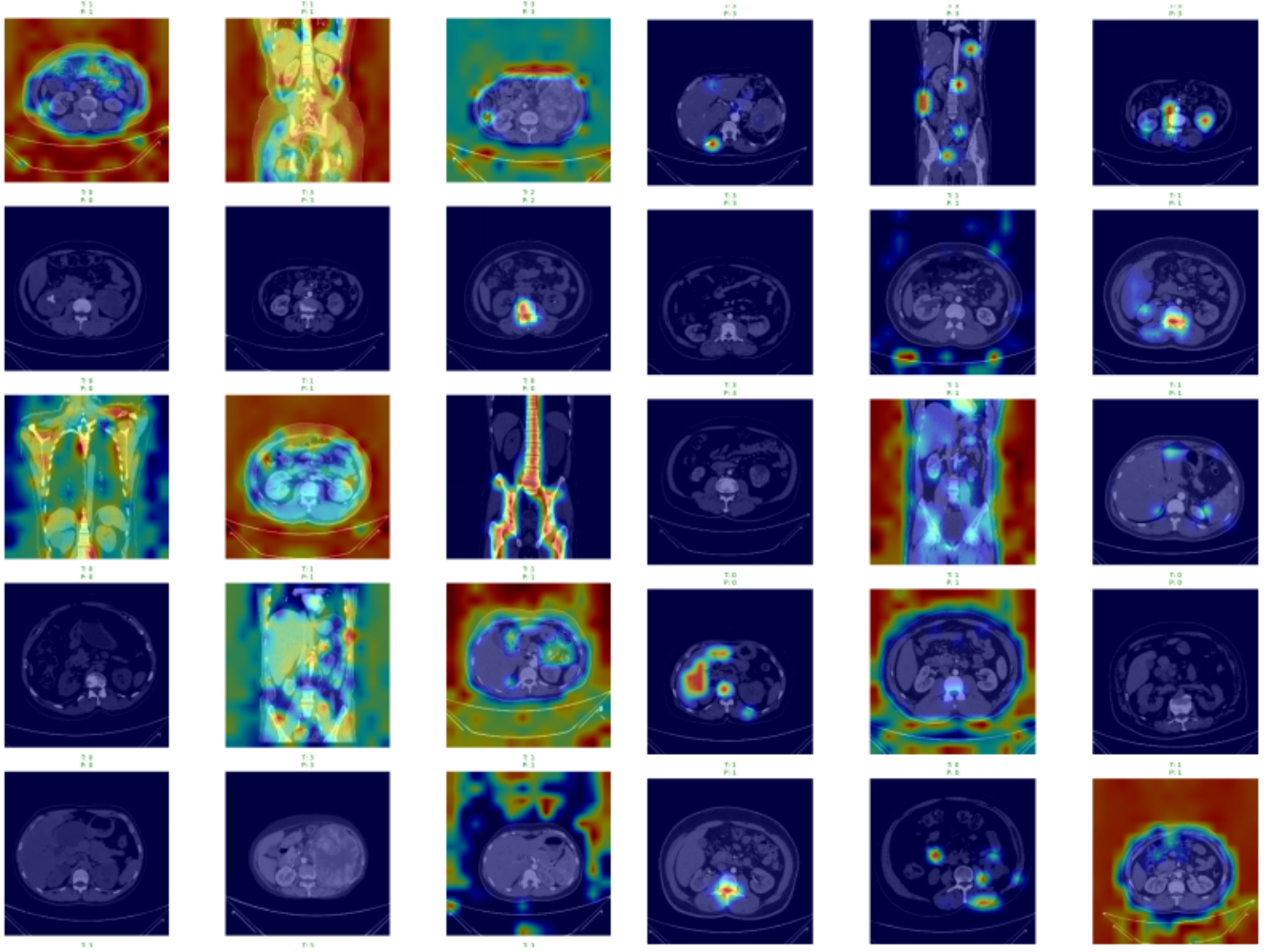


Fig. 5: DINO-based Grad-CAM [24] highlighting key regions in kidney CT images.

CT scans from four diagnostic categories—kidney cyst, normal kidney, kidney stone, and tumor—were subjected to Grad-CAM [24] analysis. Feature maps, which preserve spatial information essential for localization tasks, were taken from the last convolutional layers of CNN-based models like VGG16 and ResNet50. Grad-CAM [24] was modified to function on token-based representations because transformer-based models do not rely on convolutional feature maps. In particular, DINO-based self-supervised feature representations were used to improve the attention-weighted embeddings from ViT, resulting in more globally consistent and semantically meaningful attention maps.

The gradients are calculated mathematically with respect to the feature map activations A^k given a target class score y^c :

$$\frac{\partial y^c}{\partial A_{ij}^k} \quad (8)$$

The sensitivity of the class score to each spatial location in the feature maps is captured by these gradients. Global average pooling over the spatial dimensions is then used to determine the relevance weights for each feature map:

$$w_k^c = \frac{1}{Z} \sum_i \sum_j \frac{\partial y^c}{\partial A_{ij}^k} \quad (9)$$

where the total number of spatial places is indicated by Z . Each feature map's contribution to the final forecast is represented by these weights. Next, a weighted mixture of the feature maps and a ReLU activation are used to calculate the Grad-CAM [24] heatmap:

$$L_{\text{Grad-CAM [24]}}^c = \text{ReLU} \left(\sum_k w_k^c A^k \right) \quad (10)$$

where the feature map activations are represented by A^k and their related significance weights are indicated by w_k^c . Only features that have a positive impact on the class prediction are kept when the ReLU function is used.

The attention maps that are produced from token representations for transformer-based architectures are upsampled and modified to fit the original input image’s spatial resolution. To create comprehensible heatmaps, these maps are then superimposed on the matching CT scans.

The visuals that are produced offer a clear understanding of the areas that have an impact on the model’s choices. This makes it possible to confirm that the models concentrate on anatomical elements that are clinically useful, including kidney regions and diseased areas, rather than unimportant background aspects. As a result, this enhances the suggested system’s clinical applicability, dependability, and transparency.

IV. RESULT ANALYSIS

To guarantee an objective evaluation of generalization potential, the performance of all six models was assessed using test data that had not yet been observed. Accuracy, precision (positive predictive value), recall (sensitivity), F1-score, and Area Under the Receiver Operating Characteristic Curve (AUC) were among the evaluation criteria used. In a medical setting, where both false positives and false negatives have important consequences, these measures offer a thorough insight of model performance.

Ten-fold cross-validation was used to improve the evaluation’s robustness and dependability. This method divided the dataset into 10 subgroups, each of which was used once as a test set and the remaining subsets for training. In order to reduce variation and mitigate the impacts of data bias, the final performance metrics were produced by averaging the results across all folds.

All models’ quantitative performance is compiled in Table 2, which also offers a comparison of various assessment indi-

cators. The normalized confusion matrices and ROC curves are shown in Figures 10 and 12, respectively, providing additional information about model discrimination capacity and classification behavior.

The distribution of accurate and inaccurate predictions for each of the four classes—tumor, cyst, normal, and stone—is shown by the confusion matrices. These matrices show how well each model can differentiate between classes and identify possible patterns of misclassification. The ROC curves, on the other hand, assess the trade-off between true positive rate and false positive rate across different decision thresholds, with the AUC acting as a summary indicator of classification performance.

Figure 4 (Grad-CAM [24] pictures) displays Grad-CAM [24] visualizations for VGG16, ResNet50, ViT, and DINO-based models. By emphasizing the regions of interest utilized during classification, these visualizations offer qualitative proof of the interpretability of the model. It has been noted that CNN-based models frequently capture low-level properties by concentrating on larger, less accurate regions. On the other hand, more focused and semantically significant attention maps are generated by transformer-based models, especially ViT and DINO-enhanced architectures.

This enhanced localization suggests that transformer-based models are better at finding clinically significant patterns in CT images and capturing global contextual linkages. This capacity is further improved by the incorporation of DINO, which offers reliable feature representations acquired by self-supervised learning.

Overall, transformer-based methods—particularly ViT and DINO-based models—outperform conventional VGG [22] in both classification accuracy and interpretability, as shown by the combination of quantitative metrics and qualitative visualization. They are therefore very well suited for implementation in practical clinical decision support systems.

TABLE V: Measures of performance for the six models studied in the research.

Models	Accuracy	Class	Precision (PPV)	Recall (Sensitivity)	F1 Score
EANet [21]	98.35%	Cyst	0.96	0.98	0.97
		Normal	0.99	0.98	0.98
		Stone	1.00	0.84	0.91
		Tumor	0.92	1.00	0.96
Swin Transformer [19]	97.27%	Cyst	0.94	0.99	0.97
		Normal	1.00	0.99	0.99
		Stone	0.98	0.86	0.92
		Tumor	0.96	0.98	0.97
DINO	98.00%	Cyst	1.00	0.99	0.99
		Normal	1.00	0.95	0.98
		Stone	0.96	1.00	0.98
		Tumor	0.92	1.00	0.96
VGG16	98.47%	Cyst	0.98	0.99	0.98
		Normal	0.97	0.99	0.98
		Stone	0.98	0.97	0.97
		Tumor	1.00	0.95	0.97
		Cvst	1.00	1.00	1.00

To assess the interpretability of the suggested models, the Grad-CAM [24] analysis was performed on CT images that represented the kidney cyst, normal, stone, and tumor classes. Activation maps were taken from the last convolutional layers of CNN-based models (VGG16 and ResNet50). Token-level feature embeddings were used to modify Grad-CAM [24] for transformer-based models, such as Vision Transformer (ViT) and DINO-based representations.

Figure 4 (Grad-CAM [24] pictures) shows Grad-CAM [24] visualizations for the ViT model in the first row, the DINO-based model in the second row, and the ResNet50 and VGG16 findings in the following rows. The four classes—cyst, normal, stone, and tumor—are represented by the columns.

The visualization results show that, in comparison to CNN-based models, transformer-based models—in particular, ViT and DINO—focus more specifically on clinically significant kidney areas. ResNet50, on the other hand, frequently focuses on larger and less pertinent regions, which occasionally results in incorrect classifications.

Table 2 shows that the Vision Transformer (ViT) outperformed all other models with the highest overall accuracy of 99.37%. With an accuracy of 98.00%, the DINO-based model likewise performed well, followed by VGG16 (98.47%) and EANet [21] (98.35%). With an accuracy of 94.02%, ResNet50 performed relatively poorly.

Superior recall values were attained by the ViT model in every class, demonstrating its efficacy in reducing false negatives—a crucial aspect of medical diagnosis. In particular, it attained good memory for the normal and stone classes and

nearly flawless recall for the cyst and tumor classes.

With nearly flawless scores in every category, the precision numbers demonstrate the ViT model's resilience. Strong precision and recall were also shown by the DINO-based model, supporting its efficacy in feature representation and interpretability.

The F1-scores, which achieve values near 1.0 across all classes and show a solid balance between precision and recall, validate the ViT model's overall superiority. Among CNN-based methods, VGG16 also performed competitively, although ResNet50 fared comparatively worse because of less efficient feature localization.

It can be seen from the training curves that transformer-based models outperform CNN-based models in terms of convergence speed and stability. Specifically, ViT exhibits little overfitting and consistent learning behavior.

The Area Under the Curve (AUC) for ViT and DINO-based models approaches 1.0 across all classes, demonstrating high discriminative capability, which is further supported by the ROC curve analysis. While ResNet50 displays relatively lower numbers, VGG16 likewise demonstrates good AUC performance.

Overall, the findings show that transformer-based architectures—particularly Vision Transformer (ViT) and DINO-enhanced models—perform noticeably better in renal disease categorization than conventional CNN-based methods. Their superior interpretability and capacity to gather global contextual information make them ideal for medical imaging applications.

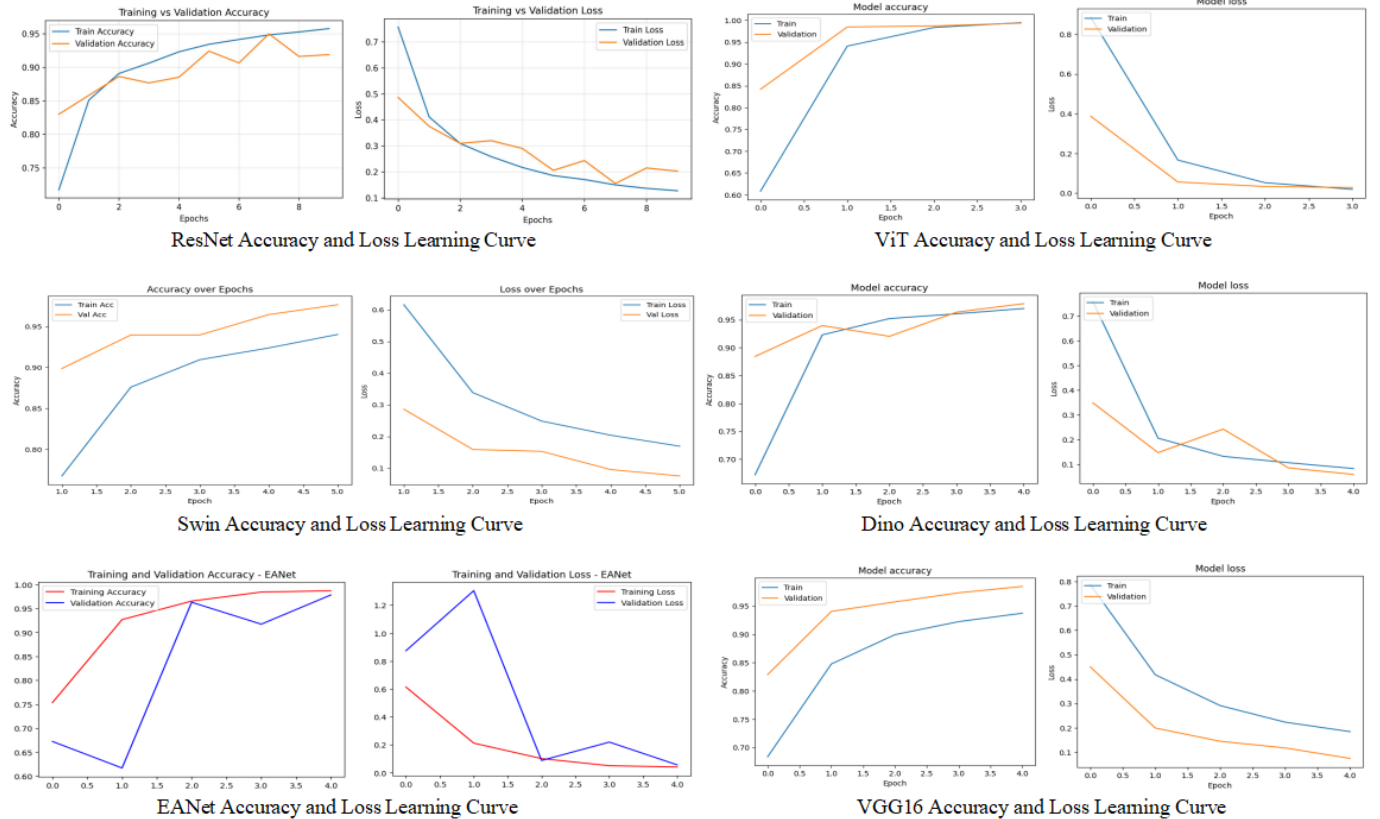


Fig. 6: Training and validation accuracy and loss curves for all models

A. Confusion matrix

To assess the effectiveness and interpretability of the model, we examined the Grad-CAM [24] visualizations and confusion matrices after randomly choosing representative pictures from various classes derived from CT scans.

For the cyst class, it is found that ResNet50 tends to pay attention to more dispersed regions, showing dependence on lower-level characteristics, whereas VGG16 concentrates on a relatively confined region (high-level features) to produce predictions. This distinction emphasizes how conventional CNN-based models are limited in their ability to capture global contextual information.

The Vision Transformer (ViT), on the other hand, shows extremely accurate categorization with few misclassifications in every class. It can successfully capture global anatomical features because to its capacity to model long-range dependencies. In a similar vein, the DINO-based model performs well, using self-supervised representations to concentrate on kidney areas that are clinically significant. VGG16 demonstrates a respectable localization of the region of interest for the stone class; nevertheless, ViT and DINO models offer more accurate and reliable attention maps, resulting in better classification performance. Conversely, ResNet50 exhibits increased misclassification rates and rather poor localization.

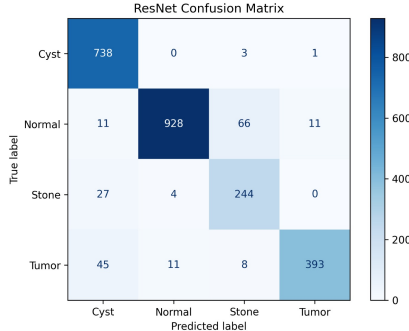


Fig. 7: Confusion matrix for the trained ResNet50 model

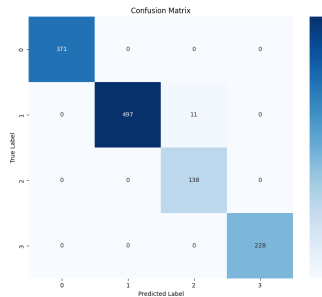


Fig. 8: Confusion matrix for the trained Vision Transformer (ViT) model

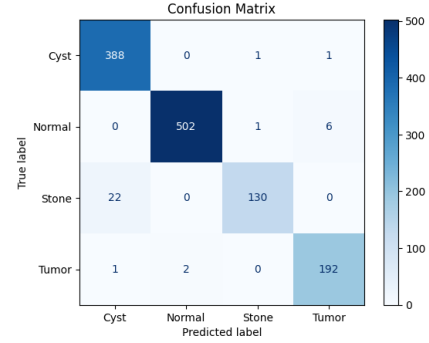


Fig. 9: Confusion matrix for the trained Swin model

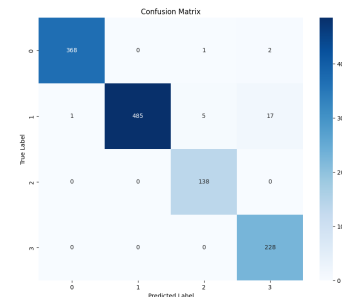


Fig. 10: Confusion matrix for the trained DINO-based model

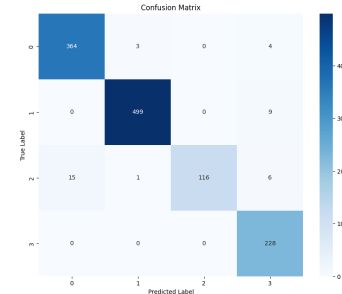


Fig. 11: Confusion matrix for the trained EANet model

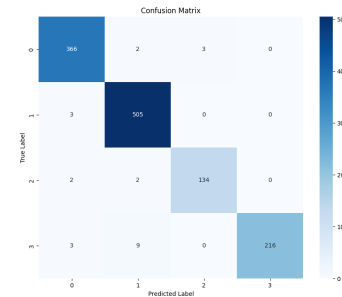


Fig. 12: Confusion matrix for the trained VGG16 model

Tumor and normal classes show a similar pattern. While ResNet50 exhibits greater misunderstanding between normal

and pathological situations, VGG16 does a respectable job of identifying pertinent regions. On the other hand, because ViT and DINO models can capture global contextual linkages, they considerably reduce such misclassifications.

In general, transformer-based methods—in particular, ViT and DINO—show better interpretability and classification performance than traditional CNN-based models.

V. CONCLUSION

- 12,446 CT scan images of the entire abdomen and urogram, including cyst, tumor, normal, and stone classes, were collected and categorized. The mean color distribution was similar across all classes, according to exploratory data analysis, suggesting that basic statistical characteristics are not enough for precise classification.
- In order to classify renal illness, this work created and assessed a number of deep learning models, including CNN-based architectures (VGG16 and ResNet50), transformer-based methods (Vision Transformer (ViT) and a DINO-based model), and an attention-based model (EANet [21]).
- The Vision Transformer (ViT) outperformed all other models, with the greatest classification accuracy of 99.37%. Additionally, the DINO-based model showed excellent performance with high accuracy and enhanced feature representation via self-supervised learning.
- Transformer-based models perform noticeably better than CNN-based models when precision, recall, and F1-score are compared. ViT is particularly good at gathering global contextual information, which is essential for identifying intricate medical patterns in CT scans.
- ViT and DINO models concentrate on clinically significant kidney areas, increasing model transparency and reliability, according to the Grad-CAM [24]-based interpretability analysis. CNN-based models, such as ResNet50, on the other hand, show poorer localization and greater misclassification rates.
- VGG16 outperforms ResNet50 among CNN-based models, but it still falls short of transformer-based methods in terms of accuracy and interpretability.
- Overall, the suggested framework shows that DINO-based models and Vision Transformer (ViT) offer a very dependable and efficient method for classifying renal diseases. They are promising instruments for practical clinical use because of their exceptional performance and explainability, which may help radiologists make early diagnoses and lower diagnostic mistakes.

REFERENCES

- [1] V. Jha et al., “Chronic kidney disease: global dimension and perspectives,” *The Lancet*, vol. 382, no. 9888, pp. 260–272, 2013.
- [2] T. Alelign and B. Petros, “Kidney stone disease: an update on current concepts,” *Advances in Urology*, 2018.
- [3] J. J. Hsieh et al., “Renal cell carcinoma,” *Nature Reviews Disease Primers*, vol. 3, 2017.
- [4] K. C. Saw et al., “Helical CT of urinary calculi,” *American Journal of Roentgenology*, vol. 175, no. 2, pp. 329–332, 2000.
- [5] Q. Zheng et al., “Computer-aided diagnosis using ultrasound imaging,” *Journal of Pediatric Urology*, 2019.
- [6] A. Parakh et al., “Urinary stone detection on CT images using deep CNNs,” *Radiology: AI*, 2019.
- [7] K. Yildirim et al., “Deep learning for kidney stone detection,” *Computers in Biology and Medicine*, 2021.
- [8] H. Zhang et al., “Kidney lesion detection using CNN,” *IEEE Access*, 2019.
- [9] N. Blau et al., “Automatic detection of renal cysts,” *Int. J. Computer Assisted Radiology*, 2018.
- [10] S. Sudharson and P. Kokil, “Ensemble deep learning for kidney ultrasound,” *Computer Methods Biomedicine*, 2020.
- [11] X. Fu et al., “Deep-learning-based CT imaging for kidney diseases,” *Journal of Healthcare Engineering*, 2021.
- [12] J. Verma et al., “Kidney stone classification using KNN and SVM,” *Pattern Recognition and Image Analysis*, 2017.
- [13] M. Islam, “CT kidney dataset,” *Kaggle*, 2021.
- [14] K. He et al., “Deep residual learning (ResNet),” *Proc. IEEE CVPR*, 2016.
- [15] F. Chollet, “Xception: Depthwise separable CNN,” *Proc. IEEE CVPR*, 2017.
- [16] M. Tan and Q. Le, “EfficientNet,” *Proc. ICML*, 2019.
- [17] A. Dosovitskiy et al., “Vision Transformer (ViT),” *arXiv:2010.11929*, 2020.
- [18] M.-H. Guo et al., “External Attention (EANet),” *arXiv:2105.02358*, 2021.
- [19] K. Simonyan and A. Zisserman, “VGG16,” *arXiv:1409.1556*, 2014.
- [20] G. Huang et al., “Stochastic depth,” *ECCV*, 2016.
- [21] R. R. Selvaraju et al., “Grad-CAM [24],” *Proc. IEEE ICCV*, 2017.
- [22] S. Patro and K. K. Sahu, “normalization [25] techniques,” *arXiv*, 2015.
- [23] Sozio S. M. et al., “Global nephrology workforce study,” *Kidney International*, 2021.
- [24] Gunasekara T. et al., “Renal biomarkers study,” *Scientific Reports*, 2022.
- [25] Bi Y. et al., “Renal carcinoma treatment analysis,” *Scientific Reports*, 2022.